

Real-Time Collaborative Assessment and Adaptive Learning Platform

Final End-to-End Product Workflow
Working reference for development

1. Product Goal

Build a shared virtual exam room where students communicate before an assessment, take a synchronized exam together, receive individual and class-level performance analysis, and use course-grounded AI to practice weak areas.

Core loop: Chat → Exam → Submission → Scoring → Comparison → Weakness Analysis → Targeted Practice → Improvement

2. Complete User Workflow

STEP	USER ACTION	SYSTEM WORK
1	Register / Login	Authenticate user and load courses/rooms.
2	Create Course	Creator creates course with university, program, semester and subject.
3	Create Room	Creator creates room linked to course and receives room code.
4	Join Room	Students enter room code and become members.
5	Pre-Exam Chat	Students chat normally. Socket.IO handles live delivery; MongoDB stores history.
6	Upload Materials	Creator uploads syllabus, notes, PDFs, previous questions and other material.
7	Process Materials	System extracts text and prepares material for retrieval.
8	Get University Pattern	System records relevant official university information and builds an exam blueprint.
9	Generate Exam	AI generates configured question types using blueprint + course material.
10	Review Exam	Creator reviews/edits generated exam before publishing.
11	Start Exam	Creator presses Start. Room changes from chat mode to live exam.
12	Live Exam	Students receive synchronized exam state, questions and server-controlled timing.
13	Submit Answers	Answers are persisted; duplicate submissions are controlled.
14	End Exam	Session ends; unfinished attempts can be auto-submitted.
15	Calculate Results	Score, accuracy, correct/incorrect/unanswered and question results are calculated.
16	Class Analysis	Class average, topic performance and question statistics are calculated.
17	Personal Analysis	Student receives strengths, weaknesses and comparison with class.
18	RAG Explanation	Relevant course material is retrieved for grounded explanations.
19	Targeted Rematch	AI generates focused practice for weak topics.
20	Improvement	Student completes rematch and compares new performance.

3. High-Level System Flow

USER → AUTH → COURSE → ROOM → REAL-TIME CHAT → MATERIALS → UNIVERSITY BLUEPRINT → AI EXAM GENERATION → EXAM REVIEW → LIVE EXAM → ATTEMPTS/ANSWERS → SCORING → CLASS ANALYTICS → PERSONAL ANALYSIS → RAG → REMATCH

4. Room State Flow

WAITING → EXAM → COMPLETED → CLOSED
WAITING: students chat and prepare. EXAM: synchronized assessment is active. COMPLETED: results/discussion/analysis are available. CLOSED: room is no longer active.

5. Exam Lifecycle

EXAM DRAFT → READY → PUBLISHED → RUNNING → COMPLETED

An Exam defines the assessment. An ExamSession represents one actual live execution.

6. Real-Time Architecture

Students ↔ Socket.IO Server ↔ Exam Engine ↔ MongoDB

The server is authoritative for exam timing, state and final submission. Typical events: student joined, exam started, answer submitted, timer state, exam completed, leaderboard updated and reconnect.

7. AI / University / RAG Flow

OFFICIAL UNIVERSITY SOURCE → EXAM BLUEPRINT

UPLOADED COURSE MATERIAL → KNOWLEDGE BASE / RAG

BLUEPRINT + COURSE MATERIAL → AI QUESTION GENERATION

STUDENT PERFORMANCE + RAG → PERSONAL EXPLANATION / TARGETED PRACTICE

8. Assessment Evaluation Flow

MCQ: exact answer matching. SAQ / 5-mark / longer answers: initially support controlled/manual evaluation; later add keyword, semantic or LLM-assisted evaluation with a rubric. Do not make AI grading the foundation of version 1.

9. Post-Exam Flow

SUBMISSION → SCORE → QUESTION ANALYSIS → TOPIC ANALYSIS → CLASS COMPARISON → PERSONAL REPORT →

RAG EXPLANATION → WEAK-TOPIC REMATCH

10. Development Workflow

PHASE	BUILD
1	Project structure + User model + authentication
2	Course + Enrollment
3	Room + RoomMembership
4	Real-time room chat
5	Exam + Question
6	ExamSession + ExamAttempt + Answer
7	Basic scoring + result page
8	Study material upload + processing
9	University source + ExamBlueprint
10	AI exam generation
11	RAG retrieval + grounded explanations
12	Class analytics + personal performance analysis
13	Targeted rematch / adaptive practice
14	Testing + security + Docker + deployment

11. MVP Boundary

Version 1 stops at authentication, course/room creation, joining, real-time chat, exam creation, MCQ execution, synchronized timing, answer submission, scoring and a basic result page. AI generation, university-source processing, RAG and adaptive rematch come after this is stable.

12. Non-Negotiable Rules

- Follow the existing User model/project conventions.
- Do not use one giant Course/Room document.
- Do not use MongoDB as the real-time transport layer.
- Do not trust the browser as the source of truth for exam timing/final state.
- Do not add Redis until needed.
- Do not add AI/RAG before the basic assessment workflow works.
- Every technology must have a concrete reason.
- Keep important code understandable for interviews.

Project North Star: Students enter a shared room, learn together, take a synchronized assessment, understand their performance relative to the class, and receive grounded practice for the topics they need to improve.